



Fall 2025 Project Proposal

Adaptive Cruise Controller

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Abstract

Adaptive Cruise Control (ACC) is an automotive feature that allows a vehicle's cruise control system to adapt the vehicle's speed to the traffic environment. A radar system attached to the front of the vehicle is used to detect whether slower moving vehicles are in the ACC vehicle's path. If a slower moving vehicle is detected, the ACC system will slow the vehicle down and control the clearance, or time gap, between the ACC vehicle and the forward vehicle. If the system detects that the forward vehicle is no longer in the ACC vehicle's path, the ACC system will accelerate the vehicle back to its set cruise control speed. This operation allows the ACC vehicle to autonomously slow down and speed up with traffic without intervention from the driver. To place this ACC into practice, two vehicles will be designed from scratch, where the ACC can be implemented in a STM32 Nucleo Boards.

Research Motivation and Goals

This project is a continuation for ACC that was simulated in Simulink but with the goal of placing classical control theory into practice. The microprocessor makes theory come into practice and can help improve safety in the streets. One of the roles I would like to explore is to test the effectiveness of Finite State Machine (FSM) or Real Time Operating Systems. This project will demonstrate realistic constraints of the ACC and will further dive the hardware aspects of both vehicles.

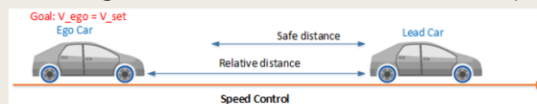
Introduction

The concept of ACC is shown below, where the leading vehicle will have a set velocity and motion whereas the ego vehicle is behind trying to maintain a set speed but will slow down soon as it approaches into the safe distance boundaries. A fixed safe distance can be determined by testing the ACC, and by considering how much acceleration the ego vehicle can imply when slowing down, since ideally the driver body cannot handle a high jerk (a quick change in acceleration when slowing down.). The track will be one dimensional to easily focus on the testing the ACC and hardware implementation run.

Modes of Operation:

The following rules are used to determine the ACC system operation mode:

- If $D_{rel} \geq D_{safe}$, then speed control mode is active. The control gain is to track the driver-set velocity, V_{set}



- If $D_{rel} < D_{safe}$, then spacing control mode is active. The control gain is to maintain the safe distance, D_{safe}

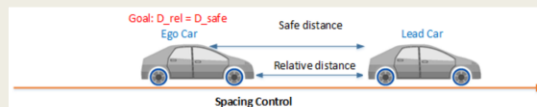


Figure 1: The vehicle setup and parameters of interest.

The ACC implementation was done through Simulink over five years ago as shown below, along with the response of both leading and ego car signals.

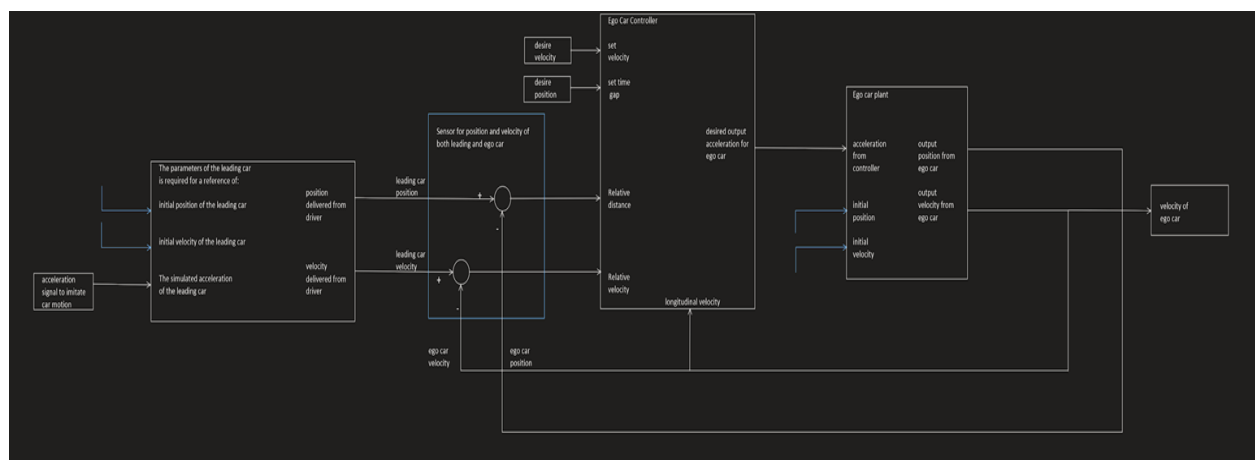


Figure 2: Block Diagram for ACC implementation.

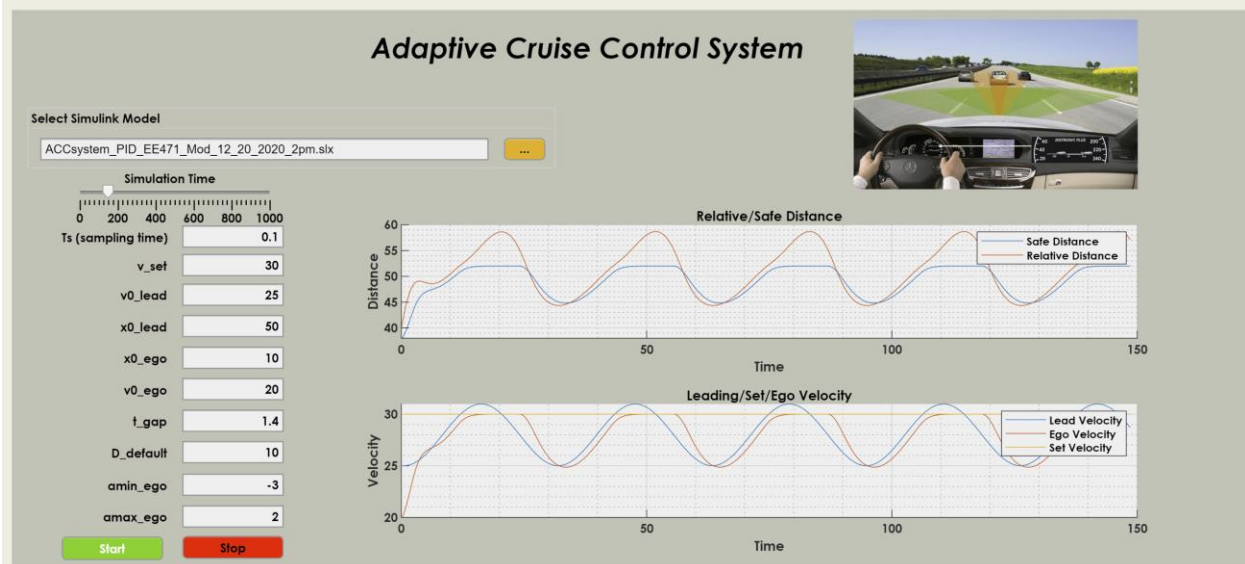


Figure 3: Desire response of leading and ego vehicle.

The hardware that is desire to implement the ACC will be Nucleo Boards from ST, and will log the signals to a computer so that signals can be interpreted and decide how to tune the ACC. The FSM algorithm can be implemented into the Nucleo Board in order to execute the ACC otherwise ideally RTOS would be heavily recommended for executing task at a set deadline.

Requirements

Hardware

- Nucleo boards
- Motor Drivers
- Chais
- Wheels
- Accelemeters sensors
- RF module or Bluetooth sensor

Software

- STM Cube IDE to program nucleo boards
- Pycharm(Python IDE) to process the signals to determine how tune ACC. Will hook up a NucleoBoard to read log the data over using UART protocol.

Timeline

Task	Deadline
Test the Nucleo Board with sensors that will be used in the project	May-July
Build the chasis of lead and ego vehicle, along with implementing the hardware.	July-August
Test the if vehicles and sensors are functional	August
Implement the ACC into the Ego vehicle, and set different motion types of leading vehicle	August-September
Design a user interface GUI using PyQt5 to log and analyze the vehicle's signal.	September
Try to implement RTOS if possible, otherwise will wrap up the report and analyses of ACC	October
Presentation	November

References

- [1] Y. Yang and O. O. Solomon, "Research on Adaptive Cruise Control Systems and Performance Analysis Using Matlab and Carsim," *School of Mechanical Engineering and Automation, Northeastern University, Shenyang, China.*